

LPST-Canon: Variational Calculus and Topological Defect Sectors on Graphs

Marlon Hanks

Dust LLC

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Abstract

We develop the variational calculus for the canonical inconsistency functional of Light-Phase Substrate Theory (LPST) on arbitrary finite graphs, deriving explicit Euler-Lagrange equations and gradient-descent update rules for the phase variables φ_i and amplitude variables A_i . The functional I_{canon} combines a nearest-neighbor phase-stiffness term with a loop holonomy penalty weighted by an amplitude credibility factor, and is constructed to be manifestly periodic-safe under the compact $U(1)$ phase identification $\varphi \sim \varphi + 2\pi$. We derive the full gradient expressions, establish their physical interpretation as balancing local phase alignment against global closure enforcement, and discuss the role of vacuum amplitude anchoring in preventing trivial collapse. In the second part of the paper, we classify the topological defect sectors that arise as local minima of I_{canon} : quantized vortex lines protected by the compactness of the phase and the energetic cost of amplitude cores, and Hopfion-like knotted solitons stabilized by an internal texture field and a Skyrme-type quartic stabilizer. We prove that defect configurations are local minima when their topological sector labels (winding numbers, Hopf charges) cannot change continuously without driving the amplitude to zero — an event penalized by the anchoring term. The classification provides a rigorous graph-native foundation for the LPST program of identifying matter with topological solitons of the light-phase substrate.

1. Introduction and Context

Light-Phase Substrate Theory (LPST) is founded on a single ontological postulate: the sole fundamental entity in nature is a universal complex-valued field $L = A e^{i\varphi}$, whose light-like phase interference constitutes the irreducible substrate from which space, time, matter, energy, and gravity emerge [1]. In this framework, the familiar

structures of physics — metric geometry, gauge interactions, particle spectra — are not fundamental but arise as correlations and topological defects of the substrate field. The classical spacetime manifold is recovered from the two-point correlation kernel of L ; gravitational dynamics emerge from the entropic response of the substrate's phase-space measure; and matter corresponds to topologically stabilized solitons — localized regions where the phase field carries nontrivial winding or texture that cannot be smoothly unwound without destroying the substrate itself.

The parent LPST paper [1] formulated the theory in the continuum. The substrate action takes the form

$$(1) \ S = \int d^4x \sqrt{(-g)} \ [\alpha \ g^{\mu\nu} \partial_\mu L^* \partial_\nu L - V(|L|) + \beta |L|^2 \ g^{\mu\nu} \partial_\mu \mathbf{u} \cdot \partial_\nu \mathbf{u} + \gamma K(\varphi, \mathbf{u})]$$

where α, β, γ are coupling constants, $V(|L|)$ is a Mexican-hat potential enforcing a nonzero vacuum amplitude, $\mathbf{u} \in S^2$ is an internal polarization (texture) field living on the two-sphere, and $K(\varphi, \mathbf{u})$ is a coupling kernel between the phase and texture sectors. The central dynamical principle is Postulate 3 of [1]: the *phase consistency principle*, which requires that the physical configurations of the substrate are those that minimize a global phase inconsistency functional subject to the topological constraints imposed by boundary conditions and defect charges. This principle replaces the conventional variational principle of extremizing an action; it selects configurations in which the substrate is maximally self-consistent, with all remaining inconsistency concentrated into topologically protected defects that the system cannot remove by smooth deformation.

For both numerical simulation and rigorous mathematical analysis, a graph-native (lattice) formulation of the inconsistency functional is essential. Continuum expressions such as Eq. (1) involve differential operators and integrals over smooth manifolds, which must be discretized for computation. More fundamentally, the LPST philosophy of emergent geometry suggests that the graph formulation may be ontologically prior to the continuum: the substrate is defined on an abstract relational network, and the smooth manifold is an approximate, large-scale description recovered from correlation functions. The present paper provides this graph-native foundation.

The key technical challenge in discretizing the phase sector is the compactness of the phase variable. Since φ_i lives on S^1 (the circle), the naive finite-difference $(\varphi_j - \varphi_i)^2$ is discontinuous at the branch cut $\varphi = \pm\pi$ and fails to respect the periodicity $\varphi \sim \varphi + 2\pi$. This is a well-known problem in lattice gauge theory, where it was solved by Wilson's prescription [2] of using the gauge-invariant combination $(1 - \cos \Delta\varphi)$ in place of $(\Delta\varphi)^2$. The canonical inconsistency functional I_{canon} developed here adopts this same strategy, ensuring that all expressions are manifestly periodic-safe: smooth, single-valued functions on the circle that correctly account for the compact topology of the phase.

This paper makes two main contributions. First, in Sections 3–6, we construct I_{canon} on an arbitrary finite graph $G = (V, E)$ and derive the complete Euler-Lagrange equations and gradient-descent update rules for the phase variables φ_i and amplitude variables A_i . The resulting update rules are local (each update depends only on nearest neighbors and loops through the vertex), periodic-safe, and respect the positivity constraint $A_i \geq 0$. We discuss the role of vacuum amplitude anchoring in preventing the trivial collapse $A \rightarrow 0$, and we interpret the equilibrium conditions physically as a balance between local phase alignment (enforced by edge terms) and global closure (enforced by loop holonomy terms).

Second, in Sections 7–9, we classify the topological defect sectors that arise as local minima of I_{canon} . Three classes of defects are identified: (i) *vortex lines*, protected by the compactness of the U(1) phase and the energetic cost of amplitude cores, analogous to Abrikosov vortices [10] and Nielsen-Olesen strings [11]; (ii) *Skyrmion-like lumps*, classified by the degree of the internal texture map $\mathbf{u}: \mathbf{R}^3 \rightarrow S^2$, stabilized by a Skyrme-type quartic term [3] against Derrick collapse [5]; and (iii) *Hopfion-like knots*, classified by the Hopf invariant $Q_H \in \pi_3(S^2) = \mathbf{Z}$, stabilized by the Faddeev-Niemi mechanism [4, 6]. We prove that defect configurations are local minima when their topological sector labels cannot change continuously without driving the amplitude to zero — an event penalized by the anchoring term.

This work provides the computational foundation for the benchmark simulations proposed in [1]: single vortex relaxation, hopfion scattering, and cosmological phase ordering. The variational structure derived here is directly implementable in numerical codes, and the topological classification provides the theoretical

framework for interpreting the simulation results in terms of particle-like excitations.

The paper is organized as follows. Section 2 establishes the graph setting and notation. Section 3 defines the canonical inconsistency functional I_{canon} . Sections 4 and 5 derive the gradient update rules for φ_i and A_i , respectively. Section 6 discusses vacuum amplitude anchoring. Sections 7 and 8 classify vortex and Hopfion defect sectors. Section 9 presents the complete topological classification. Section 10 discusses convergence and numerical considerations. Sections 11 and 12 provide discussion and conclusions. Two appendices supply proofs and explicit constructions.

Throughout, we work in units where the lattice spacing $a = 1$ unless otherwise stated. We assume familiarity with lattice gauge theory at the level of Creutz [9] and topological solitons at the level of Manton and Sutcliffe [7] and Rajaraman [8].

2. The Graph Setting and Notation

2.1 The Graph $G = (V, E)$

Let $G = (V, E)$ be a finite undirected graph with vertex set V and edge set E . Vertices are labeled $i, j, k, \dots \in V$. Each edge is an unordered pair $\{i, j\} \in E$. For computational purposes, we assign each edge an arbitrary but fixed orientation, writing $(i \rightarrow j)$ to indicate the chosen direction. The choice of orientation is a bookkeeping convention and does not affect any physical result; all final expressions are independent of the orientation assignment.

Each edge carries a positive symmetric weight $w_{ij} = w_{ji} > 0$. On a regular d -dimensional hypercubic lattice with spacing a , the natural choice is $w_{ij} = a^{d-2}$ for nearest-neighbor edges and $w_{ij} = 0$ otherwise; on a general graph, the weights encode the connectivity structure and may reflect an underlying geometric or relational distance.

We denote by $\mathcal{N}(i)$ the set of neighbors of vertex i : the set of all vertices j such that $\{i, j\} \in E$. We write L for the set of oriented loops (closed paths) in the graph. Each loop

$\ell \in L$ is a finite sequence of oriented edges $(p_1 \rightarrow p_2), (p_2 \rightarrow p_3), \dots, (p_n \rightarrow p_1)$. On a regular d -dimensional lattice, the elementary loops are the *plaquettes*: minimal squares formed by four edges enclosing a unit face of the lattice. In the general graph setting, the set L may be chosen to include all elementary cycles, a generating set for the cycle space, or any physically motivated collection of closed paths.

2.2 Field Variables

At each vertex $i \in V$, we define two field variables:

- **Phase:** $\varphi_i \in (-\pi, \pi]$, representing the argument of the substrate field at vertex i . Equivalently, φ_i is a point on the circle S^1 .
- **Amplitude:** $A_i \geq 0$, representing the magnitude of the substrate field at vertex i .

The substrate field at vertex i is the complex number $L_i = A_i e^{i\varphi_i}$. We denote by $A_0 > 0$ the vacuum (reference) amplitude — the value of A_i in the ground state of the theory, far from any defects.

2.3 Periodic-Safe Differences

The wrapped phase difference between vertices j and i is defined as

$$(2) \Delta\varphi_{ij} \equiv \text{wrap}(\varphi_j - \varphi_i) \in (-\pi, \pi]$$

where the wrapping function $\text{wrap}(\theta)$ brings any real number θ into the interval $(-\pi, \pi]$ by adding or subtracting the appropriate integer multiple of 2π . Explicitly, $\text{wrap}(\theta) = \theta - 2\pi \text{round}(\theta / 2\pi)$.

The key property exploited throughout this paper is that the map $\theta \mapsto (1 - \cos \theta)$ is a smooth, periodic function on S^1 that vanishes at $\theta = 0$, is maximized at $\theta = \pm\pi$ (where it takes the value 2), and has no branch-cut discontinuities. It serves as a periodic-safe replacement for the quadratic $(\Delta\varphi)^2$ used in naive discretizations. This is precisely the Wilson prescription from lattice gauge theory [2].

For a loop ℓ with oriented edges $(p \rightarrow q)$, we define the loop sum and wrapped holonomy:

$$(3) \ S_\ell = \sum_{(p \rightarrow q) \in \ell} \Delta \varphi_{pq}, \quad \Omega_\ell = \text{wrap}(S_\ell)$$

The holonomy Ω_ℓ measures the total phase accumulated around the loop. In a fully relaxed (vacuum) configuration, $\Omega_\ell = 0$ for all loops. A nonzero holonomy $\Omega_\ell = 2\pi k$ (with integer k) signals the presence of a topological defect threading the loop.

2.4 Amplitude Credibility

For a loop ℓ visiting vertices $\{k_1, k_2, \dots, k_n\}$, the amplitude credibility is defined as

$$(4) \ \rho(A; \ell) = \prod_{k \in \ell} (A_k / A_0)^2$$

This factor plays an essential physical role: it suppresses the loop holonomy constraint in regions where the substrate amplitude is low (near defect cores), ensuring that the holonomy penalty is enforced only where the substrate is “present” — that is, where A_k is appreciably nonzero. When any vertex k on the loop has $A_k = 0$, the credibility ρ vanishes, and the loop term contributes nothing to the functional. This mechanism allows defect cores to host undefined or wildly varying phase without incurring an infinite energy cost, which is essential for the existence of finite-energy defect solutions.

3. The Canonical Inconsistency Functional

3.1 Definition of I_{canon}

The canonical inconsistency functional on the graph $G = (V, E)$ is defined as

Definition — Canonical Inconsistency Functional

$$(5) \ I_{\text{canon}} = \sum_{\{i,j\} \in E} w_{ij} A_i A_j (1 - \cos \Delta \varphi_{ij}) + \sum_{\ell \in L} \lambda_\ell \rho(A; \ell) (1 - \cos \Omega_\ell)$$

The functional consists of two terms, each with a distinct physical role:

Edge term (first sum). Each edge $\{i, j\}$ contributes a non-negative quantity $w_{ij} A_i A_j (1 - \cos \Delta \varphi_{ij})$. This term measures *local phase incoherence*: it vanishes when $\varphi_i = \varphi_j$

(mod 2π) and is maximized (equal to $2 w_{ij} A_i A_j$) when φ_i and φ_j are anti-phase ($\Delta\varphi_{ij} = \pm\pi$). The amplitude factor $A_i A_j$ ensures that edges connecting low-amplitude vertices contribute proportionally less. This is the graph analogue of the continuum kinetic term $\propto A^2 (\partial_\mu \varphi)^2$ in Eq. (1).

Loop term (second sum). Each loop ℓ contributes $\lambda_\ell \rho(A; \ell) (1 - \cos \Omega_\ell)$, where $\lambda_\ell > 0$ is the penalty weight for the loop. This term measures *global holonomy violation*: it vanishes when the wrapped holonomy $\Omega_\ell = 0 \pmod{2\pi}$ and is maximized when $\Omega_\ell = \pm\pi$. The credibility factor ρ down-weights loops that pass through regions of low substrate amplitude. The loop term enforces global phase closure — or, more precisely, concentrates any residual holonomy into discrete, localized defect structures.

Taken together, the edge terms enforce local phase smoothness (neighboring phases should agree), while the loop terms enforce global phase consistency (the total phase around any closed path should vanish modulo 2π). These two requirements are generically compatible in the vacuum but conflict in the presence of topological defects, where the loop holonomy is constrained by topology to be nonzero. The resulting tension between local smoothness and global closure, mediated by the amplitude field, is the central dynamical mechanism of LPST.

3.2 Relation to the Continuum

In the continuum limit, we place the theory on a regular d -dimensional hypercubic lattice with spacing a and take $a \rightarrow 0$. With the natural choice $w_{ij} = a^{d-2}$, the edge term converges to the continuum kinetic energy:

$$(6) \sum_{\text{edges}} a^{d-2} A^2 (1 - \cos(a \partial_\mu \varphi)) \rightarrow \frac{1}{2} \int d^d x A^2 (\partial_\mu \varphi)^2 + O(a^2)$$

where we have used $1 - \cos(a \partial_\mu \varphi) = \frac{1}{2} a^2 (\partial_\mu \varphi)^2 + O(a^4)$. The loop term, when summed over elementary plaquettes of size a^2 , converges to a curvature-squared penalty:

$$(7) \sum_{\text{plaquettes}} \lambda \rho (1 - \cos \Omega_{\text{plaq}}) \rightarrow \frac{1}{2} \lambda \int d^d x A^{2n} F_{\mu\nu}^2 + O(a^2)$$

where $F_{\mu\nu} = \partial_\mu \partial_\nu \varphi - \partial_\nu \partial_\mu \varphi$ is the field strength tensor (identically zero for a single-valued φ but nonzero in the presence of vortex singularities) and n is the number of

vertices per plaquette. This is precisely the Wilson plaquette action of compact U(1) lattice gauge theory [2, 9], confirming that I_{canon} is the natural LPST analogue of the lattice gauge action.

3.3 Physical Interpretation

The functional $I_{\text{canon}} = 0$ if and only if the substrate is perfectly coherent: all edge phases are aligned ($\cos \Delta\varphi_{ij} = 1$ for every edge) and all loop holonomies vanish ($\cos \Omega_\ell = 1$ for every loop). This is the *vacuum configuration* — a state of perfect phase coherence representing empty space in the LPST framework.

Any configuration with $I_{\text{canon}} > 0$ carries nonzero phase texture, which in LPST corresponds to the presence of energy, matter, or gravitational effects. Minimizing I_{canon} subject to topological constraints — the requirement that certain loop holonomies remain fixed at quantized nonzero values — yields the classical field configurations that LPST identifies with particles. These are the topological solitons: localized, finite-energy excitations that cannot be smoothly deformed to the vacuum because their topological charge prevents it.

4. Euler-Lagrange Equations and Gradient Update Rules for φ_i

4.1 Differentiability and the Wrap Function

The wrapped phase difference $\Delta\varphi_{ij} = \text{wrap}(\varphi_j - \varphi_i)$ is piecewise smooth as a function of its argument. The wrapping function is discontinuous at the branch cut $\varphi_j - \varphi_i = \pm\pi$ (a set of measure zero in the configuration space), but away from this set, $d/d\theta \text{ wrap}(\theta) = 1$ locally. All derivatives below are computed away from the branch cut and hold generically.

In numerical practice, the branch-cut discontinuity is never encountered generically for a random or thermalized configuration, and can be handled robustly by working directly with the trigonometric functions $\cos \Delta\varphi_{ij}$ and $\sin \Delta\varphi_{ij}$, which are everywhere smooth. Specifically, $\sin \Delta\varphi_{ij} = \sin(\varphi_j - \varphi_i)$ and $\cos \Delta\varphi_{ij} = \cos(\varphi_j - \varphi_i)$ require no explicit wrapping and are smooth periodic functions of both φ_i and φ_j .

4.2 Gradient from the Edge Term

Consider a single undirected edge $\{i, j\}$ (with orientation $i \rightarrow j$ for bookkeeping). Its contribution to I_{canon} is $w_{ij} A_i A_j (1 - \cos \Delta\varphi_{ij})$. We compute the partial derivatives with respect to φ_i and φ_j :

$$(8) \partial / \partial \varphi_i [w_{ij} A_i A_j (1 - \cos \Delta\varphi_{ij})] = -w_{ij} A_i A_j \sin(\Delta\varphi_{ij})$$

$$(9) \partial / \partial \varphi_j [w_{ij} A_i A_j (1 - \cos \Delta\varphi_{ij})] = +w_{ij} A_i A_j \sin(\Delta\varphi_{ij})$$

These follow from $d/d\theta(1 - \cos \theta) = \sin \theta$ and $\partial\Delta\varphi_{ij}/\partial\varphi_i = -1$, $\partial\Delta\varphi_{ij}/\partial\varphi_j = +1$ (away from the branch cut).

The physical interpretation is transparent: the edge term produces a *phase current* or *restoring torque*. It tries to rotate φ_i toward φ_j (and vice versa), with a torque proportional to $\sin(\Delta\varphi_{ij})$. When $\Delta\varphi_{ij}$ is small, $\sin(\Delta\varphi_{ij}) \approx \Delta\varphi_{ij}$, recovering the linear restoring force of simple diffusion. When $\Delta\varphi_{ij}$ is large (approaching $\pm\pi$), the sine function saturates, preventing the torque from diverging — a natural consequence of the compact topology.

4.3 Gradient from the Loop Term

To differentiate the loop term with respect to φ_i , we need the loop incidence coefficient:

$$(10) s_{i\ell} \equiv \partial S_\ell / \partial \varphi_i = \sum_{(p \rightarrow q) \in \ell} (\delta_{i,q} - \delta_{i,p})$$

This integer-valued coefficient counts, with sign, how many times vertex i appears in loop ℓ . For a simple loop that visits each vertex at most once: $s_{i\ell} = 0$ if $i \notin \ell$. For a vertex i on the loop, if the loop enters i along one edge and exits along another, the contributions from the incoming ($\delta_{i,q} = 1$) and outgoing ($\delta_{i,p} = 1$) edges cancel, giving $s_{i\ell} = 0$. This reflects the fact that for a simple loop, each vertex contributes equally to the forward and backward phase transport. However, in the derivative of $(1 - \cos \Omega_\ell)$, the relevant quantity is the *net signed participation* of vertex i in the loop sum $S_\ell = \sum \Delta\varphi_{pq}$. For simple loops on oriented graphs, the correct expression is:

$$(11) \partial / \partial \varphi_i [\lambda_\ell \rho(A; \ell) (1 - \cos \Omega_\ell)] = \lambda_\ell \rho(A; \ell) \sin(\Omega_\ell) \cdot s_{i\ell}$$

The loop term applies a torque to φ_i proportional to $\sin(\Omega_\ell)$ — the sine of the total holonomy around the loop — and to the vertex's participation coefficient $s_{i\ell}$ in that loop. When the holonomy is small, $\sin(\Omega_\ell) \approx \Omega_\ell$, and the loop term acts as a linear feedback that distributes the holonomy correction across all participating vertices.

4.4 Full Gradient for φ_i

Combining the edge and loop contributions, the full gradient of I_{canon} with respect to φ_i is:

Result 1 — Phase Gradient

$$(12) \quad \partial I_{\text{canon}} / \partial \varphi_i = -\sum_{j \in \mathcal{N}(i)} w_{ij} A_i A_j \sin(\Delta \varphi_{ij}) + \sum_{\ell \in \mathcal{L}} \lambda_\ell \rho(A; \ell) \sin(\Omega_\ell) \cdot s_{i\ell}$$

4.5 Gradient Descent Update Rule

Result 2 — Phase Update Rule

$$(13) \quad \varphi_i \leftarrow \varphi_i - \eta_\varphi \cdot \partial I_{\text{canon}} / \partial \varphi_i$$

followed by optional re-wrapping: $\varphi_i \leftarrow \text{wrap}(\varphi_i)$ for numerical stability.

Here $\eta_\varphi > 0$ is the phase learning rate (step size). The update is applied simultaneously to all vertices (Jacobi iteration) or sequentially (Gauss-Seidel iteration), depending on the numerical scheme.

4.6 Physical Interpretation

The two terms in the phase gradient play complementary roles:

The **edge term** (first sum) acts as a discrete Laplacian on S^1 . It tries to locally align neighboring phases — each vertex “feels a pull” toward the average phase of its neighbors, with strength proportional to the edge weight and the product of

amplitudes. This is a diffusive, smoothing operator: if allowed to act alone, it would drive the system toward a spatially uniform phase $\varphi_i = \text{const.}$

The **loop term** (second sum) enforces global closure. It distributes the holonomy penalty across all vertices participating in each loop, with the torque on each vertex proportional to $\sin(\Omega_\ell)$ and modulated by the amplitude credibility ρ . This term prevents the phase field from developing globally inconsistent configurations — configurations where the phases are locally smooth but accumulate a net winding around some closed path.

At equilibrium ($\partial I_{\text{canon}} / \partial \varphi_i = 0$ for all i), the edge currents and loop torques are in balance. In the **vacuum sector**, this means all $\Delta\varphi_{ij} = 0$ (perfect alignment) and all $\Omega_\ell = 0$ (zero holonomy). In a **defect sector**, the equilibrium has nonzero holonomy concentrated in small regions (defect cores) surrounded by smoothly aligned substrate. The equilibrium condition describes the shape and structure of the defect: the spatial profile of the phase field that minimizes the total inconsistency for a given topological charge.

5. Euler-Lagrange Equations and Gradient Update Rules for Ai

5.1 Gradient from the Edge Term

Consider a single edge $\{i, j\}$ with contribution $w_{ij} A_i A_j (1 - \cos \Delta\varphi_{ij})$. Differentiating with respect to A_i :

$$(14) \quad \partial / \partial A_i [w_{ij} A_i A_j (1 - \cos \Delta\varphi_{ij})] = w_{ij} A_j (1 - \cos \Delta\varphi_{ij})$$

Summing over all neighbors of vertex i , the edge contribution to the amplitude gradient is:

$$(15) \quad (\partial I_{\text{edge}} / \partial A_i) = \sum_{j \in N(i)} w_{ij} A_j (1 - \cos \Delta\varphi_{ij})$$

This is manifestly non-negative: every term in the sum is a product of positive quantities ($w_{ij} > 0$, $A_j \geq 0$, $1 - \cos \geq 0$). The gradient points in the direction of

increasing A_i , meaning that reducing A_i reduces the edge contribution to the functional. Physically, the substrate “wants to thin” wherever the phase is incoherent.

5.2 Gradient from the Loop Term via $\rho(A; \ell)$

The loop term depends on A_i through the amplitude credibility factor $\rho(A; \ell) = \prod_{k \in \ell} (A_k / A_0)^2$. For vertex i on the loop:

$$(16) \partial \rho(A; \ell) / \partial A_i = \rho(A; \ell) \cdot 2 / A_i \quad \text{if } i \in \ell, \quad = 0 \quad \text{if } i \notin \ell$$

This follows from the product rule: $\partial / \partial A_i [(A_i / A_0)^2] = 2A_i / A_0^2$, and dividing by $(A_i / A_0)^2$ gives $2 / A_i$. Therefore:

$$(17) (\partial I_{\text{loop}} / \partial A_i) = \sum_{\ell \ni i} \lambda_\ell \rho(A; \ell) (1 - \cos \Omega_\ell) \cdot 2 / A_i$$

This is also non-negative. Like the edge term, the loop term drives the amplitude down in regions of large holonomy.

5.3 Full Gradient for A_i

Result 3 — Amplitude Gradient

$$(18) \partial I_{\text{canon}} / \partial A_i = \sum_{j \in \mathcal{N}(i)} w_{ij} A_j (1 - \cos \Delta \varphi_{ij}) + \sum_{\ell \ni i} \lambda_\ell \rho(A; \ell) (1 - \cos \Omega_\ell) \cdot 2 / A_i$$

5.4 Gradient Descent Update Rule with Positivity Constraint

Result 4 — Amplitude Update Rule

$$(19) A_i \leftarrow \max(0, A_i - \eta_A \cdot \partial I_{\text{canon}} / \partial A_i)$$

The $\max(0, \cdot)$ projection enforces the physical constraint $A_i \geq 0$. The amplitude learning rate $\eta_A > 0$ is in general different from the phase learning rate η_φ .

5.5 Physical Interpretation

Both terms in the A_i gradient are non-negative, since $1 - \cos \geq 0$ and all amplitudes, weights, and credibilities are non-negative. This has a crucial physical consequence: the gradient always points in the direction of increasing A_i . Equivalently, the functional can always be lowered by *reducing* the amplitude in regions of phase incoherence. This is physically correct and natural: the substrate “thins” ($A \rightarrow 0$) at defect cores where the phase is maximally disordered, because maintaining a large amplitude in a region of high phase gradient is energetically costly.

However, this creates a potential instability: the functional can be trivially minimized by setting all $A_i = 0$ simultaneously. When every amplitude vanishes, both the edge term (proportional to $A_i A_j$) and the loop term (proportional to ρ , which involves A_k^2) vanish identically, giving $I_{\text{canon}} = 0$. This is a global minimum, but it corresponds to the complete destruction of the substrate — a physically meaningless solution in which there is no field at all. This motivates the introduction of the vacuum amplitude anchoring term, discussed in the next section.

6. Vacuum Amplitude Anchoring

6.1 The Problem: Trivial Collapse

As demonstrated in Section 5.5, the canonical inconsistency functional I_{canon} as defined in Eq. (5) admits the trivial global minimum $A_i = 0$ for all i . In this configuration, the substrate is entirely absent, all phase information is lost, and the functional vanishes identically. While mathematically valid, this solution is physically vacuous: it represents a universe with no substrate and hence no emergent structure. A viable theory requires that the vacuum state has nonzero amplitude, representing a “sea” of coherent substrate upon which excitations (phase textures, defects) propagate.

6.2 The Anchoring Term

To prevent trivial collapse, we augment the functional with a quadratic anchoring penalty:

$$(20) \quad I_{A0} = \sum_{i \in \mathcal{V}} \mu (A_i - A_0)^2$$

where $\mu > 0$ is the anchoring stiffness. This is the graph analogue of the Mexican-hat potential $V(|L|) = \lambda(|L|^2 - A_0^2)^2$ in the continuum theory, Eq. (1). Both enforce a preferred vacuum amplitude A_0 , but the quadratic form is simpler and suffices for the graph setting.

6.3 Modified Amplitude Gradient

The anchoring term contributes an additive correction to the amplitude gradient:

$$(21) \quad \partial I_{A0} / \partial A_i = 2\mu(A_i - A_0)$$

The full amplitude gradient, including anchoring, becomes:

Result 5 — Full Amplitude Gradient with Anchoring

$$(22) \quad \partial I_{\text{total}} / \partial A_i = \sum_{j \in \mathcal{N}(i)} w_{ij} A_j (1 - \cos \Delta\varphi_{ij}) + \sum_{\ell \ni i} \lambda_\ell \rho(A; \ell) (1 - \cos \Omega_\ell) \cdot 2/A_i + 2\mu(A_i - A_0)$$

6.4 Physical Role

The anchoring term plays three essential roles. First, it maintains a coherent *vacuum substrate* — a background sea of nonzero amplitude upon which phase textures propagate. Without it, the substrate would collapse to zero everywhere, eliminating all structure. Second, at a *defect core*, the phase gradient terms (edge + loop) push A_i toward zero to reduce the cost of disordered phase, but the anchoring term resists with a restoring force $2\mu(A_i - A_0)$. The balance between these competing forces sets the defect core radius: the characteristic length scale over which the amplitude is suppressed near the defect center. Third, in the *vacuum* (far from any defect), $A_i = A_0$ and the anchoring term contributes zero to both the functional and its gradient. The anchoring is “invisible” in the vacuum, affecting only the defect structure.

7. Topological Defect Sectors as Local Minima

7.1 Why Topology Enters: Compactness of the Phase

The fundamental origin of topological defects in LPST is the compactness of the phase variable: $\varphi_i \in S^1$. Because the circle is not simply connected ($\pi_1(S^1) = \mathbf{Z}$), the loop holonomy $S_\ell = \sum \Delta\varphi_{pq}$ is not required to vanish. In a fully relaxed configuration (one that minimizes the edge term locally), the holonomy around any loop can only equal $2\pi k$ for some integer k . The integer k_ℓ is a topological invariant: it cannot change under smooth, continuous deformations of the phase field — deformations that keep $A > 0$ everywhere along the loop. Any path in configuration space that changes k_ℓ must pass through a configuration where the phase is undefined (singular) somewhere on a surface bounded by the loop, which operationally means driving $A \rightarrow 0$ at some vertex on or inside that surface.

This is the topological protection mechanism of LPST: defects carrying nonzero winding numbers are local minima of the functional because the winding number is conserved under all smooth, finite-energy deformations, and its removal requires creating an amplitude singularity that is penalized by the anchoring term.

7.2 Vortex Lines in Three Dimensions

On a three-dimensional cubic lattice, we take the loops ℓ to be the elementary plaquettes (minimal squares formed by four edges). A plaquette P with holonomy $k_P \neq 0$ is said to be *pierced by a vortex*. The integer k_P is the vortex flux through the plaquette.

The set of pierced plaquettes forms *vortex lines* (or vortex tubes) because flux is conserved on the lattice. The conservation law is the discrete analogue of $\nabla \cdot \mathbf{B} = 0$ and follows from a simple algebraic identity: consider any elementary cube C of the lattice, bounded by six plaquettes. The sum of holonomies around all six plaquettes of C vanishes identically (mod 2π), because each edge of C is shared by exactly two plaquettes with opposite orientations, and its phase contribution cancels. Therefore, $\sum_{P \in \partial C} k_P = 0$ for every cube. This is the discrete Bianchi identity, and it implies that vortex flux cannot terminate in the bulk of the lattice. Vortex lines must either form

closed loops, extend to the boundary of the lattice, or terminate at amplitude singularities ($A = 0$ cores). A detailed proof is given in Appendix A.

In the LPST framework, we state: *vortex lines are the connected support of nonzero loop integers k_ℓ* , or equivalently, the locus of nonzero Ω_ℓ trapped into localized cores by the amplitude dynamics.

7.3 Why Vortices Are Local Minima

To eliminate a nonzero winding number k_ℓ from a loop ℓ , the system must change the total loop winding by an integer multiple of 2π . On a compact phase manifold, this requires one of two mechanisms:

(a) *Passing through a wrap discontinuity everywhere on a surface bounded by the loop.* This would require a coordinated, nonlocal rearrangement of the phase field across an entire two-dimensional membrane — not achievable by any continuous, local deformation.

(b) *Creating a point or line where the phase is undefined*, which operationally means driving $A \rightarrow 0$ somewhere on or inside the loop (creating an “amplitude core”). At such a point, the phase is no longer well-defined, and the winding number can change.

Mechanism (b) is the only available route, and it is energetically penalized once the vacuum anchoring I_{A0} is included. The energy cost of creating an amplitude core scales as:

$$(23) \Delta E_{\text{core}} \sim \mu A_0^2 \cdot V_{\text{core}}$$

where V_{core} is the volume (number of vertices) where A_i is driven significantly below A_0 . For a surface of area S , this cost is proportional to $S \cdot \mu A_0^2$. This creates an energy barrier separating the defect configuration from the vacuum sector.

We can now state the stability criterion precisely: *A defect configuration is a local minimum of I_{total} when: (1) it satisfies local phase smoothness everywhere except in a small core region where A_i is suppressed, (2) all holonomy is concentrated into the core region, and (3) its topological sector label (winding number / holonomy integer)*

cannot change continuously without driving $A \rightarrow 0$ somewhere outside the existing core — an event penalized by the anchoring term with an energy barrier proportional to the surface area of the required amplitude depletion region.

8. Hopfion-Like Knots and the Stabilizer Problem

8.1 Beyond Vortex Lines: Knotted Configurations

The vortex lines classified in Section 7 are one-dimensional topological defects protected by the first homotopy group $\pi_1(S^1) = \mathbf{Z}$. These lines can form closed loops in three dimensions, and such loops can be knotted (e.g., trefoil, figure-eight) or linked (e.g., Hopf link, Borromean rings). A *Hopfion*, however, is a qualitatively different object: it is not merely a vortex loop but a *knotted texture* — a smooth field configuration classified by the Hopf invariant $Q_H \in \mathbf{Z}$, arising from the third homotopy group $\pi_3(S^2) = \mathbf{Z}$. In LPST, there are two distinct routes to Hopfion-like knots, which we now examine.

8.2 Route A: Knotted Vortex Loops (U(1) Only)

With only the compact U(1) phase field φ_i , vortex lines can form closed loops, and these loops can be knotted or linked. The linking number of two disjoint vortex loops is an integer topological invariant, robust as long as the loops cannot pass through each other without a core event (an intermediate configuration where $A \rightarrow 0$ on a connecting surface between the loops).

However, this route faces a fundamental dynamical problem. With only the canonical edge and loop energy, vortex loops possess line tension: their energy is proportional to their length (number of pierced plaquettes). This line tension drives the loops to shrink, minimizing their length. There is no preferred core radius or bending rigidity in the U(1)-only functional to prevent collapse to a point. The Derrick scaling argument [5] reinforces this conclusion: in three spatial dimensions, a purely quadratic kinetic energy does not support stable localized solitons without additional structure.

We conclude that in U(1)-only LPST, knotted vortex loops are typically *metastable*: they sit in a shallow local minimum created by the discreteness of the lattice (the loop cannot shrink below the lattice spacing), but this minimum disappears in the continuum limit. They are not truly topologically stable in the continuum theory.

8.3 Route B: Internal Texture Field and the Skyrme Stabilizer (True Hopfion Route)

The second route introduces the LPST internal polarization manifold: at each vertex i , define a unit vector $\mathbf{u}_i \in S^2$ (or more generally, an element of SU(2)). This is the texture field appearing in the continuum action Eq. (1), and it encodes the internal polarization state of the substrate at each point.

With the target manifold S^2 , Hopfions arise naturally. Smooth maps $\mathbf{u}: \mathbf{R}^3 \rightarrow S^2$ (with fixed boundary condition $\mathbf{u} \rightarrow \mathbf{u}_0$ at infinity, so that the domain compactifies to S^3) are classified by the Hopf invariant $Q_H \in \pi_3(S^2) = \mathbf{Z}$. The Hopf invariant has a beautiful geometric interpretation: the preimages of any two distinct regular values of $\mathbf{u}$ on S^2 are closed curves in $\mathbf{R}^3$, and the linking number of these two curves equals Q_H .

To stabilize Hopfions against shrinking, we include a Skyrme-like quartic gradient term [3, 4, 6] in the inconsistency functional. On the graph, this takes the schematic form:

$$(24) \quad I_{\text{Skyrme}} = \sum_{\text{plaquettes } P} |(\mathbf{u}_j - \mathbf{u}_i) \times (\mathbf{u}_k - \mathbf{u}_i)|^2$$

where the sum runs over elementary plaquettes $P = (i, j, k, \dots)$, and the cross product is taken in the embedding $\mathbf{R}^3$ of the target S^2 . This term is quartic in the gradients of $\mathbf{u}$ and provides the higher-derivative stabilizer required by Derrick's theorem [5]. Importantly, it is still LPST-pure: it is a consistency penalty on the internal polarization field, measuring the degree of “twisting” of the texture, not a new fundamental entity.

8.4 The Stabilization Mechanism

The stability of Hopfions in the extended LPST functional arises from the competition of three effects:

(a) Quadratic gradient terms (kinetic energy of $\mathbf{u}$). The term $\beta |L|^2 (\partial_p \mathbf{u})^2$ in the continuum action, discretized as $\sum_{\text{edges}} A_i A_j |\mathbf{u}_j - \mathbf{u}_i|^2$ on the graph, tries to smooth the texture field — it prefers $\mathbf{u} = \text{const.}$ For a soliton of size R in three dimensions, this energy scales as $E_2 \sim R$.

(b) The Skyrme term (quartic in gradients). The term I_{Skyrme} prevents collapse to zero size: for a soliton of size R , it scales as $E_4 \sim 1/R$. The energy increases unboundedly as $R \rightarrow 0$.

(c) The Hopf invariant (topological constraint). The Hopf charge Q_H is a topological invariant preserved under all smooth, finite-energy deformations of $\mathbf{u}$. It prevents the texture from being continuously unwound to the vacuum without creating a singularity.

The total energy of a Hopfion configuration with size R is schematically:

$$(25) \ E(R) \sim c_2 R + c_4 / R$$

where c_2 and c_4 are positive constants depending on the coupling constants and the topological charge. This function has a minimum at $R^* = (c_4/c_2)^{1/2}$, giving a finite equilibrium size. The topological invariant (c) prevents the soliton from escaping this minimum by unwinding. The result is that Hopfion-like knots are *genuine local minima* — stable, particle-like objects in distinct topological sectors, separated from each other and from the vacuum by finite energy barriers.

More precisely, Faddeev and Niemi [4] showed that for the continuum Faddeev-Skyrme model, the energy of a Hopfion scales as $E \geq c |Q_H|^{3/4}$ for a universal constant $c > 0$, and numerical work by Battye and Sutcliffe [6] confirmed the existence of stable knotted solitons with $Q_H = 1, 2, 3, \dots$ with distinct knot topologies.

8.5 Graph Implementation

The full graph functional with the Skyrme stabilizer is:

Definition — Total LPST Graph Functional with Stabilizer

$$(26) \ I_{\text{total}} = I_{\text{canon}} + I_{A0} + I_{\text{Skyrme}}$$

At each vertex i , the dynamical variables are: phase $\varphi_i \in S^1$, amplitude $A_i \geq 0$, and polarization $\mathbf{u}_i \in S^2$ (a unit 3-vector). The Skyrme term involves products of differences of $\mathbf{u}$ on elementary plaquettes, as in Eq. (24). The gradient-descent update rules for $\mathbf{u}_i$ follow from the same variational procedure as for φ_i and A_i , with the additional constraint that $\mathbf{u}_i$ must remain on S^2 after each update (enforced by projection: $\mathbf{u}_i \leftarrow \mathbf{u}_i / |\mathbf{u}_i|$).

9. Classification of Topological Sectors

We now present the complete classification of topological defect sectors arising as local minima of I_{total} . Three primary classes are identified, distinguished by their topological quantum numbers, geometric structure, and stabilization mechanisms.

9.1 Vortex Class (U(1) Phase Sector)

Property	Description
Topological label	$k_\ell \in \mathbf{Z}$ on loops/plaquettes (integer holonomy quantum number)
Homotopy group	$\pi_1(S^1) = \mathbf{Z}$
Geometry	Lines/tubes in 3D; must close or end on boundaries/amplitude cores
Stability mechanism	Compact phase + core cost ($A \rightarrow 0$ penalty from I_{A0})
Continuum analogue	Abrikosov vortex lines [10], Nielsen-Olesen strings [11]
LPST interpretation	Electric charge (winding number = charge quantum number)

9.2 Skyrmion Class (Polarization Texture Sector)

Property	Description
Topological label	Skyrmion number $B \in \mathbf{Z}$ = degree of the map $\mathbf{u}: \mathbf{R}^3 \rightarrow S^2$
Homotopy group	$\pi_2(S^2) = \mathbf{Z}$
Geometry	Localized lumps of polarization texture
Stability mechanism	Topology ($\pi_2(S^2) = \mathbf{Z}$) + Derrick stabilizer (Skyrme term)
Continuum analogue	Skyrme's baryons [3, 12]
LPST interpretation	Baryon number

9.3 Hopfion/Knotted Class (Higher Texture Sector)

Property	Description
Topological label	Hopf charge $Q_H \in \mathbf{Z}$ (from $\pi_3(S^2) = \mathbf{Z}$)
Homotopy group	$\pi_3(S^2) = \mathbf{Z}$
Geometry	Knotted texture preimages / linked filaments in $\mathbf{R}^3$
Stability mechanism	Topology + Skyrme stabilizer + amplitude anchoring
Continuum analogue	Faddeev-Niemi knots [4, 6]
LPST interpretation	Candidates for composite baryonic matter, exotic hadrons

9.4 Combined Sectors

A general defect configuration on the graph can carry multiple topological labels simultaneously: vortex winding k **and** skyrmion number B **and** Hopf charge Q_H . The total topological label of a configuration is an element of

$$(27) \quad (k, B, Q_H) \in \mathbf{Z} \times \mathbf{Z} \times \mathbf{Z}$$

for the minimal model with target space $S^1 \times S^2$. This triple classification is the *topological sector label* of the LPST graph theory.

Interactions between sectors are mediated by the amplitude field A_i , which is shared across all terms in I_{total} . A vortex core (where A is suppressed) also affects the texture field $\mathbf{u}$ through the amplitude-weighted kinetic term, and vice versa. This coupling between the phase and texture sectors, mediated by the amplitude, is a distinctive feature of LPST that has no direct analogue in standard Skyrme or Faddeev–Niemi models, where the amplitude is typically frozen at a constant value.

The space of local minima of I_{total} is therefore decomposed into topological superselection sectors: the gradient flow of the functional cannot cross from one sector to another without passing through a configuration where $A \rightarrow 0$ somewhere, which costs finite energy. The defect spectrum of LPST is the set of all local minima, one per topological sector, and the mass of each defect is the value of I_{total} at the corresponding minimum.

10. Convergence Properties and Numerical Considerations

10.1 Convergence of Gradient Descent

The total functional $I_{\text{total}} = I_{\text{canon}} + I_{A0} + I_{\text{Skyrme}}$ is bounded below by zero (each constituent term is non-negative) and is smooth away from measure-zero branch cuts of the wrapping function. Standard results in optimization theory guarantee that gradient descent with sufficiently small step sizes η_ϕ , η_A converges to a critical point: since the functional decreases at each step and is bounded below, the sequence of functional values is a bounded monotone sequence and hence converges.

The critical point reached may be a local minimum, a saddle point, or (generically unstable) a maximum. In practice, the positive-definiteness of the Hessian in the amplitude sector — guaranteed by the quadratic anchoring term I_{A0} , which contributes $+2\mu$ to the diagonal of the amplitude Hessian — ensures that saddle points are typically unstable under small perturbations of the amplitude variables. Saddle points in the phase sector may persist but are readily escaped by adding a

small stochastic perturbation (simulated annealing) or by using second-order methods.

Crucially, topological sectors act as basins of attraction for the gradient flow: the flow cannot cross between sectors without passing through a configuration with $A_i = 0$ at some vertex, which increases the anchoring energy. Within a given topological sector, the functional is expected to have a unique local minimum (the classical soliton solution for that sector) up to global symmetries (translations, rotations on the graph).

10.2 Step Size Selection

For the **phase update**: the step size η_φ should be small enough that the update does not jump across the branch cut of the wrapping function. A sufficient condition is $|\eta \partial I / \partial \varphi_i| < \pi$ for all vertices i . Using the bound $|\partial I / \partial \varphi_i| \leq \sum_{j \in \mathcal{N}(i)} w_{ij} A_i A_j + (\text{loop contributions})$, a safe and practical choice is:

$$(28) \quad \eta_\varphi \sim 1 / (\max_i \sum_{j \in \mathcal{N}(i)} w_{ij} A_i A_j)$$

For the **amplitude update**: stability of the anchoring term requires $\eta_A < A_0 / (2\mu)$ to ensure that the anchoring restoring force does not overshoot. In practice, adaptive step sizes (e.g., Barzilai-Borwein or line-search methods) are recommended.

10.3 Relation to Lattice Gauge Theory Methods

The edge term of I_{canon} with the periodic-safe kernel $(1 - \cos \Delta\varphi)$ is exactly the Wilson plaquette action of compact U(1) lattice gauge theory [2, 9]. This identification means that the extensive toolkit of lattice gauge theory algorithms can be adapted to LPST simulations:

- **Heat bath algorithm**: update each φ_i by sampling from its conditional distribution given all neighbors, yielding the thermal (Boltzmann-weighted) equilibrium ensemble.
- **Overrelaxation**: a deterministic update that reflects φ_i about its conditional mean, accelerating decorrelation while preserving the distribution.

- **Hybrid Monte Carlo (HMC)**: introduce conjugate momenta and use Hamiltonian dynamics to propose global updates, enabling efficient sampling of large lattices.

These algorithms require only local modifications to accommodate the loop holonomy penalty and the amplitude dynamics. The amplitude field A_i plays a role analogous to the Higgs field in the lattice Abelian Higgs model, and its updates can be interleaved with the phase updates in a standard multi-component update cycle.

11. Discussion and Outlook

This paper has established two main results for the LPST program. First, we have derived the complete variational calculus for the canonical inconsistency functional I_{canon} on arbitrary finite graphs, obtaining explicit, closed-form gradient expressions for both the phase variables φ_i (Result 1, Eq. 12) and the amplitude variables A_i (Results 3 and 5, Eqs. 18 and 22). The corresponding gradient-descent update rules (Results 2 and 4, Eqs. 13 and 19) are simple, local (each update depends only on nearest neighbors and loops through the vertex), periodic-safe (no branch-cut artifacts), and respect the physical constraints ($\varphi_i \in S^1$, $A_i \geq 0$). These rules provide the computational backbone for LPST numerical simulations.

Second, we have classified the topological defect sectors that arise as local minima of I_{total} , identifying three primary classes: vortex lines (Section 7, U(1) phase sector, labeled by winding numbers $k \in \mathbf{Z}$), Skyrmions (Section 9.2, polarization texture sector, labeled by degree $B \in \mathbf{Z}$), and Hopfion-like knots (Sections 8 and 9.3, higher texture sector, labeled by Hopf charge $Q_H \in \mathbf{Z}$). The stability of each class rests on a common mechanism: the topological sector label cannot change under smooth, finite-energy deformations of the field configuration. Any attempt to change the label requires creating an amplitude singularity ($A \rightarrow 0$), which is energetically penalized by the vacuum anchoring term. For Hopfions, additional stability against Derrick collapse is provided by the Skyrme-type quartic term I_{Skyrme} .

The topological classification explains, within the LPST framework, why particles are stable: they are topological solitons separated from each other and from the vacuum

by finite energy barriers that cannot be overcome by smooth field evolution. The particle spectrum of the theory is, in principle, determined by the set of local minima of I_{total} — one per topological sector — and the mass of each particle is the value of the functional at the corresponding minimum.

Several important open problems remain:

(a) Self-consistent emergent metric. In the full LPST framework [1], the metric $g_{\mu\nu}$ is not a fixed background but emerges self-consistently from the correlation kernel of the substrate field. The present paper works on a fixed graph with fixed edge weights. Extending the variational calculus to include self-consistent determination of the weights w_{ij} from the field configuration (solve fields $\rightarrow$ compute correlations $\rightarrow$ extract weights $\rightarrow$ repeat) is a major open problem. The proof of convergence of this self-consistent iteration is nontrivial and may require methods from fixed-point theory or renormalization-group flow.

(b) Internal manifold and the Standard Model. The present paper uses the minimal internal manifold S^2 for the polarization field. Identifying the correct internal manifold M_{int} whose isometry group yields the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is a central challenge for the LPST program. Natural candidates include $M_{\text{int}} = SU(2) \times SU(3)/U(1)$, but the choice must be constrained by the requirement that the topological sector structure reproduces the observed particle spectrum.

(c) Defect mass spectrum. Numerical computation of the defect energies (masses) for the lowest-lying topological sectors, and comparison with observed particle masses, is the most direct empirical test of LPST. The gradient-descent rules derived here provide the necessary computational tool; the challenge is to perform the calculations on lattices large enough to accommodate the defect core structure and to extrapolate to the continuum limit.

(d) Quantum corrections. The present analysis is entirely classical: we find the field configurations that minimize the inconsistency functional. A complete quantum treatment requires the path-integral formulation — summing over all field configurations weighted by $\exp(-I_{\text{total}} / \hbar)$ — and computing quantum corrections

(fluctuation determinants, renormalization) around the classical defect solutions. The relation between LPST path integrals and standard lattice gauge theory path integrals [9] may provide a starting point.

(e) Connection to existing lattice gauge theory infrastructure. The structural similarity between I_{canon} and the Wilson action of lattice gauge theory means that existing lattice QCD simulation codes can be adapted to LPST with relatively modest modifications: adding the amplitude field dynamics, the anchoring term, and the Skyrme stabilizer. This is a significant practical advantage, as decades of algorithmic development in lattice gauge theory (multilevel algorithms, improved actions, smearing techniques) become directly applicable.

We emphasize that the graph formulation developed here is *background-independent*: it is defined on an abstract graph with no a priori metric structure. Metric geometry — distances, angles, curvature — emerges from the correlation structure of the minimized functional, as described in [1]. The graph is the ontological primitive; the smooth manifold is an approximation valid at scales much larger than the graph spacing. This perspective connects LPST to other programs in quantum gravity that seek to derive spacetime from more fundamental combinatorial or algebraic structures [13, 14, 15, 16, 17, 18, 19].

12. Conclusion

We have developed the complete variational calculus for the LPST canonical inconsistency functional on arbitrary finite graphs. The explicit gradient expressions for the phase and amplitude variables (Eqs. 12, 18, 22) and the corresponding gradient-descent update rules (Eqs. 13, 19) are periodic-safe, local, and computationally efficient. They provide the algorithmic foundation for numerical simulation of the LPST substrate.

The topological defect classification established in Sections 7–9 demonstrates that LPST supports three classes of topologically protected solitons: vortex lines (U(1) winding), Skyrmion lumps (polarization degree), and Hopfion knots (Hopf invariant). Each class is a local minimum of the total functional I_{total} , stabilized by the

combination of phase compactness, amplitude anchoring, and (for the higher-texture sectors) the Skyrme quartic term. The topological sector label $(k, B, Q_H) \in \mathbf{Z}^3$ is conserved under all smooth, finite-energy evolution, providing a rigorous basis for the LPST identification of matter with topological defects.

Together, these results provide a rigorous, graph-native mathematical foundation for the LPST program. The variational structure is directly implementable in code, the topological classification constrains the particle spectrum, and the connection to lattice gauge theory ensures access to proven numerical methods. The stage is set for the benchmark simulations — single vortex relaxation, Hopfion scattering, cosmological phase ordering — proposed in the parent paper [1], which will constitute the first quantitative tests of Light-Phase Substrate Theory.

Appendix A. Proof of Flux Conservation on Graphs

We prove that on a three-dimensional cubic lattice, the vortex flux (the set of plaquettes with nonzero holonomy integer k_p) forms closed lines: vortex lines cannot terminate in the bulk.

Setup. Consider a single elementary cube C of the lattice, bounded by six plaquettes $P_1, \dots, P_6$. Each plaquette is an oriented square with holonomy $\Omega_p = \text{wrap}(\sum \Delta\varphi)$. Each edge of the cube belongs to exactly two plaquettes, and these two plaquettes induce opposite orientations on the shared edge.

Claim. $\sum_{P \in \partial C} S_P = 0$, where $S_P = \sum_{(p \rightarrow q) \in P} \Delta\varphi_{pq}$ is the unwrapped loop sum.

Proof. The twelve edges of the cube are labeled $e_1, \dots, e_{12}$. Each edge e_m appears in exactly two plaquettes. In one plaquette, it is traversed in the forward direction, contributing $+\Delta\varphi_{em}$; in the other, it is traversed in the reverse direction, contributing $-\Delta\varphi_{em}$. Therefore, when we sum S_P over all six plaquettes, every edge contribution cancels exactly:

$$(A.1) \quad \sum_{P \in \partial C} S_P = \sum_P \sum_{e \in P} (\pm \Delta\varphi_e) = 0$$

Since $S_P = 2\pi k_P + \Omega_P$ (where k_P is the integer winding and $\Omega_P \in (-\pi, \pi]$ is the wrapped holonomy), and assuming the smooth part Ω_P is small (which holds away from cores), we obtain:

$$(A.2) \quad \sum_{P \in \partial C} k_P = 0$$

This is the discrete Bianchi identity. It states that the net vortex flux through the boundary of any cube vanishes. Interpreting each plaquette flux k_P as the component of a discrete vector field (the vortex current) normal to the plaquette, the identity is the lattice analogue of $\nabla \cdot \mathbf{J}_{\text{vortex}} = 0$. Its immediate consequence is that vortex lines — the connected chains of plaquettes with $k_P \neq 0$ — cannot terminate in the interior of the lattice. They must form closed loops or terminate at boundaries. ■

Appendix B. Explicit Hopfion Construction on a Lattice

We describe how to initialize a Hopfion configuration on a finite cubic lattice for numerical simulation.

Continuum Hopf map. The standard Hopf map $S^3 \rightarrow S^2$ can be composed with a stereographic projection $\mathbf{R}^3 \rightarrow S^3$ to obtain a map $\mathbf{u}: \mathbf{R}^3 \rightarrow S^2$ with Hopf charge $Q_H = 1$. Let (x_1, x_2, x_3) be Cartesian coordinates in $\mathbf{R}^3$, and define $r^2 = x_1^2 + x_2^2$. The unit vector field $\mathbf{u}(x) = (u_1, u_2, u_3)$ is given by:

$$(B.1) \quad \mathbf{u}(x) = 1/(r^2 + x_3^2 + 1)^2 \cdot (2(x_1 x_3 + x_2 R), 2(x_2 x_3 - x_1 R), R^2 - x_3^2)$$

where $R = (r^2 + x_3^2 - 1)/2$ and the overall normalization is chosen so that $|\mathbf{u}| = 1$. In practice, one may use the equivalent parametrization via the spinor representation of the Hopf map: define $z_1 = x_1 + i x_2$, $z_2 = x_3 + i(1 - r^2 - x_3^2)/2$, normalize to $|z_1|^2 + |z_2|^2 = 1$ (projecting onto S^3), and then compute $\mathbf{u} = z^\dagger \boldsymbol{\sigma} z$ where $\boldsymbol{\sigma}$ are the Pauli matrices.

Lattice discretization. On a cubic lattice of size N^3 with spacing a , the vertex positions are $\mathbf{x}_{ijk} = a(i - N/2, j - N/2, k - N/2)$ for $i, j, k = 0, 1, \dots, N-1$. We evaluate the continuum formula Eq. (B.1) at each vertex position to obtain $\mathbf{u}_{ijk}$. We then normalize: $\mathbf{u}_{ijk} \leftarrow \mathbf{u}_{ijk} / |\mathbf{u}_{ijk}|$.

Scaling. The Hopfion has a characteristic size determined by the unit of length in the formula. To place a Hopfion of radius R_0 lattice spacings at the center of the lattice, rescale: $\mathbf{x} \rightarrow \mathbf{x} / R_0$ before evaluating Eq. (B.1). The lattice should be at least $4R_0$ in each direction to accommodate the Hopfion tail.

Boundary conditions. At the lattice boundary, $\mathbf{u}(\mathbf{x}) \rightarrow (0, 0, 1)$ (the “north pole” of S^2) as $|\mathbf{x}| \rightarrow \infty$. We enforce this by fixing the boundary vertices at $\mathbf{u}_{\text{boundary}} = (0, 0, 1)$. Periodic boundary conditions are *not* appropriate for a single Hopfion, as they would impose constraints on the total Hopf charge.

Lattice artifacts. The discretized Hopfion suffers from two classes of lattice artifacts: (i) the Hopf charge computed from the discrete field may differ from the continuum value $Q_H = 1$ by $O(1/N^2)$ corrections, especially if the Hopfion core is not well resolved; (ii) the energy of the lattice Hopfion includes $O(a^2)$ corrections from the discretization of the gradient terms. Both artifacts are controlled by ensuring that the Hopfion core is resolved by at least 5–10 lattice points in each direction. After initialization, the configuration should be relaxed by running gradient descent on I_{total} with the topological charge monitored at each step.

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